Errata to "Basis construction for polynomial spline spaces over arbitrary T-meshes"

The article [1] contains errors and omissions, which are corrected as follows. Page numbers refer to the published version.

1. **Page 5, Eqs. (2) and (3); page 9, Theorem 2.** Replace $(d_1 + 1 + c_h)(d_2 + 1 + c_v)$ with $(d_1 + 1 + c_v)(d_2 + 1 + c_h)$ in Eqs. (2) and (3). In Theorem 2, replace the definition of α with

$$\alpha = (d_1 + 1 + c'_v)(d_2 + 1 + c'_h).$$

2. **Page 5, Definition 6 and Example 3.** In Definition 6, replace "with one segment coinciding with l" with "with one horizontal segment contained in l"; the vertical case is analogous. In Example 3, replace the T l-edge v_4v_5 with v_1v_5.

3. **Page 7, Lemma 2.** Replace $0 \le s < r$ with $0 \le s \le r$.

4. **Page 8, first paragraph of Example 6.** Replace both occurrences of l_2 with l_1, and replace "the blue dashed line" with "the green dashed line".

5. **Pages 9–10, proof of Theorem 2.** Where the denominator exponent is omitted, including in Eq. (10), replace $\partial_{\pm}^{d_2}/\partial y$ with $\partial_{\pm}^{d_2}/\partial y^{d_2}$.

6. **Page 11, paragraph following Definition 8.** Replace $\mathcal{B}_i \in \mathcal{P}_{d_1,d_2}$ with $\mathcal{B}_i|_C \in \mathcal{P}_{d_1,d_2}$ for every cell C of $\mathrm{ext}_s(\mathcal{T})$.

7. **Pages 11–12, first step of the proof of Theorem 3.** Replace "for each virtual edge e_i, $i = 1, \ldots, s$, in $\mathcal{T}$" with "for each extended edge e_i, $i = 1, \ldots, s$, in $\mathrm{ext}_s(\mathcal{T})$ but not in $\mathcal{T}$". In the two subsequent derivative identities, replace all four restrictions $|_{y=y_k}$ and $|_{x=x_k}$ with $|_{e_i}$.

8. **Page 13, last sentence of Example 8.** Replace the sentence beginning "These shared B-splines result" with:

   > The shared B-splines in Fig. 8(b) couple the two sets of EEE conditions, which may therefore need to be solved jointly.

9. **Pages 14–15, Proposition 1 and its proof.** Add the following qualification to the statement:

   > The EEE solution basis is chosen so that the resulting spline basis retains the standard nonnegative tensor-product B-spline basis of the tensor-product part of the original mesh $\mathcal{T}$.

   Replace the proof with:

   > Write $f_j = \sum_{i=1}^{n} \widetilde{c}_{i,j}\mathcal{B}_i$ and, after reordering, denote the retained tensor-product functions by $f_1, \ldots, f_m$. They are nonnegative and satisfy $\sum_{i=1}^{m} f_i = 1$. For $j > m$, choose $a_j \geq \max\{0, -\min_\Omega f_j\}$ and set
   >
   > $$g_i = f_i \quad (1 \leq i \leq m), \qquad g_j = f_j + a_j \sum_{i=1}^{m} f_i \quad (m < j \leq h).$$
   >
   > Then all g_j are nonnegative. The change-of-basis matrix has nonnegative entries and is upper triangular with unit diagonal. It is therefore invertible, and $g_1, \ldots, g_h$ form a nonnegative basis.

10. **Page 17, Example 10, following the second linear system.** Replace "The solution space is empty" with "The solution space is $\{(0, 0)\}$".

11. **Page 18, Example 11, following the five linear equations.** Replace the expression for f with
    $$f = \frac{36}{3245}B_1 - \frac{648}{3245}B_2 - \frac{387}{649}B_3 + B_4.$$
    In the following sentence, replace the knot vector of B_0, $[1, 2, 6, 7] \times [1, 2, 6, 7]$, with $[0, 1, 2, 6] \times [1, 2, 6, 7]$. The final basis remains $f + B_0$ together with a PT-spline basis for $\mathcal{T}_1$.

12. **Page 19, Table 4.** In the column $\{3\}\times[2, 3]$, interchange the polynomial factors $-\frac{3}{4}y^2+4y-\frac{9}{2}$ and $-\frac{1}{6}y^2 + \frac{7}{6}y - \frac{17}{12}$ in the B_1 and B_2 entries. In the B_3 right-hand derivative entry of the column $\{3\} \times [1, 2]$, replace $0 \times (x - 1)^2/3$ with $0 \times (y - 1)^2/3$.

13. **Page 19, Example 12, T *l*-edges component.** Replace the EEE condition with
    $$-\frac{4}{25}\left(\frac{x-2}{2}\right)^4 c_1 + \frac{4}{27}\left(\frac{x-2}{2}\right)^4 c_2 \equiv 0.$$
    Replace both occurrences of 972/7925 with 27/25.

14. **Page 21, Conclusion and Acknowledgement.** In the Conclusion, replace "LR B-splines lack linear independence" with "LR B-splines are not guaranteed to be linearly independent". In the Acknowledgement, replace "T-cycles (Example 14)" with "T-cycles (Example 11)".

15. **Pages 1–2 and 21, citation typographical errors.** On page 1, replace "Yin, 20004" with "Yin, 2004". On page 2, replace "Sederberg et aal." with "Sederberg et al.", "LI and Scott" with "Li and Scott", and "De Boor and De Boor" with "de Boor". On page 21, remove the duplicated author from the reference to *A Practical Guide to Splines*.

References

[1] S. Zhong, B. Huang, F. Chen, Basis construction for polynomial spline spaces over arbitrary T-meshes, *Computer Aided Geometric Design* **126** (2026), 102534. doi:10.1016/j.cagd.2026.102534.